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Place and Sort 4 DoF Robot Manipulator in a Warehouse

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Chapter 1

INTRODUCTION

In current warehouse management, efficacy along with accuracy is essential to raising production levels and achieving some of the organizational goals. As a result, robotic automation has become a considerable asset in the achievement of such goals, with manipulators being highly utilized in sorting, organizing, and transporting goods. This project focuses on the design and implementation of a 4 Degree of Freedom (DoF) robot manipulator for sorting and placing objects within a warehouse.



Figure 1.1: Robotic Manipulator in action

The four Degrees of Freedom robot manipulator can accomplish a number of intricate functions requiring precision and some spatial manipulation of objects thus ideal for warehouse automation. It also has an articulated arm that eases motion in different planes enabling the robotic part to pick up, carry and place objects in

specific angles. There is also adaptability of the robot, decreasing the necessity of manual work in the performing such actions that are repetitive thereby increasing both the sorting speed and accuracy.

In this project, the robot manipulator is able to identify the items, classify them according to the rules of the given system and store the items in the appropriate place. With the use of the sensors and control algorithms, the robot can move and perform tasks even in a changing environment as found in a warehouse. Because these processes would be carried out by the system, the system aims to enhance control of the inventories, manage the processes effectively, and hence increase the efficiency of warehouses operations.

This report also explains the process of designing, system construction and enunciating functions of the 4 DoF robot manipulator and describes, in general, how such a system can alter the concept of warehousing systems.

Chapter 2

DH-Convention

2.1 Denavit-Hartenberg (DH) Convention

The Denavit-Hartenberg (DH) convention is a systematic method used in robotics to describe the kinematic structure of a robot, particularly the relationships between consecutive joints. It simplifies the process of deriving forward kinematics, which involves calculating the position and orientation of the robot's end-effector relative to its base.

The DH convention uses a set of four parameters to describe each joint in a robot manipulator:

- **Link Length (a):** The distance between the axes of two consecutive joints along the common normal.
- **Link Twist (α):** The angle between the two consecutive joint axes.
- **Link Offset (d):** The distance along the previous joint axis to the common normal.
- **Joint Angle (θ):** The rotation around the current joint axis to align with the next.

By applying these parameters, the DH convention defines a transformation matrix for each joint, making it easier to model and analyze the overall motion of robotic manipulators.

2.2 Coordinate Frame Assignment

The image shows a 4 Degree of Freedom (DoF) robotic arm with multiple coordinate frames assigned at different joints, each corresponding to a part of the robot's

kinematic structure. The coordinate systems are labeled "Coordinate System 1," "Coordinate System 2," "Coordinate System 4," and so on. These frames are crucial for describing the position and orientation of each link relative to its preceding link, following the Denavit-Hartenberg (DH) convention.

- **Coordinate System 1** appears to be attached near the end-effector, showing the frame for the final link, which likely defines the position of the gripper or tool.
- **Coordinate System 2** is placed at the base of the robot, representing the reference frame from which all subsequent joint transformations originate.
- **Coordinate System 4** is located at the second joint, indicating the rotation axis of the arm where joint transformations will be applied.

Each frame defines the origin and orientation of the respective link, with the X, Y, and Z axes labeled to indicate directions in 3D space. The consistency in frame assignment is crucial for calculating forward and inverse kinematics, as it allows for the precise determination of the robot's end-effector position through sequential transformations across joints.

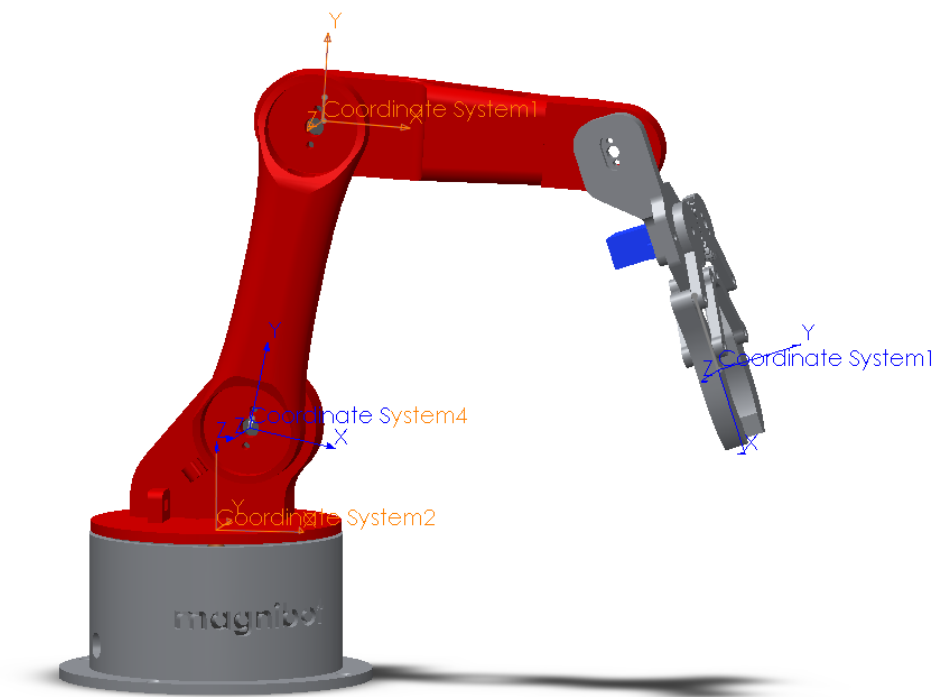


Figure 2.1: Coordinate Frame Assignment

2.3 DH-Convention of the Robot

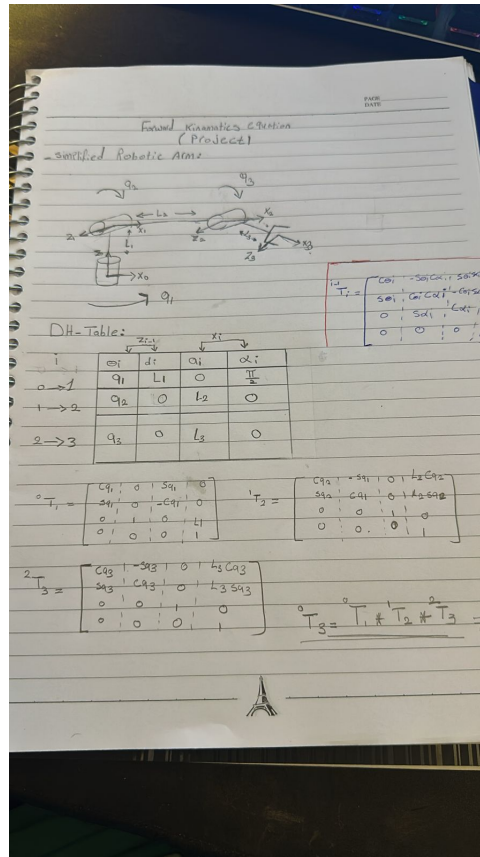


Figure 2.2: DH-convention by hand

2.3.1 DH-Parameter Table

The Denavit-Hartenberg (DH) table specifies the key parameters needed to describe the transformations between consecutive links of a robotic manipulator. These parameters capture the relative positions and orientations of adjacent joints, enabling efficient kinematic modeling.

i	θ_i	d_i	a_i	α_i
1	$\theta_1 = q_1$	L_1	0	$\frac{\pi}{2}$
2	$\theta_2 = q_2$	0	L_2	0
3	$\theta_3 = q_3$	0	L_3	0

Table 2.1: Denavit-Hartenberg Parameters

2.3.2 Transformation Matrices

Based on the DH-Table, the individual transformation matrices between the frames can be defined as follows:

- **Transformation matrix from frame 0 to frame 1 (0T_1):**

$$\begin{aligned}
 &{}^0T_1 \\
 &= \\
 &\begin{bmatrix} \cos(q_1) & -\sin(q_1) & 0 & 0 \\ \sin(q_1) & \cos(q_1) & 0 & 0 \\ 0 & 0 & 1 & L_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{aligned} \tag{2.1}$$

- **Transformation matrix from frame 1 to frame 2 (1T_2):**

$$\begin{aligned}
 &{}^1T_2 \\
 &= \\
 &\begin{bmatrix} \cos(q_2) & -\sin(q_2) & 0 & L_2 \cos(q_2) \\ \sin(q_2) & \cos(q_2) & 0 & L_2 \sin(q_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{aligned} \tag{2.2}$$

- **Transformation matrix from frame 2 to frame 3 (2T_3):**

$$\begin{aligned}
 &{}^2T_3 \\
 &= \\
 &\begin{bmatrix} \cos(q_3) & -\sin(q_3) & 0 & L_3 \cos(q_3) \\ \sin(q_3) & \cos(q_3) & 0 & L_3 \sin(q_3) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{aligned} \tag{2.3}$$

- **Final Transformation Matrix (0T_3):**

$${}^0T_3 = {}^0T_1 \times {}^1T_2 \times {}^2T_3 \tag{2.4}$$

- In the Matrix form:

$${}^0T_3 = \begin{bmatrix} \cos(q_1 + q_2 + q_3) & -\sin(q_1 + q_2 + q_3) & 0 & L_1 + L_2 \cos(q_2) + L_3 \cos(q_3) \\ \sin(q_1 + q_2 + q_3) & \cos(q_1 + q_2 + q_3) & 0 & L_2 \sin(q_2) + L_3 \sin(q_3) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2.5)$$

2.4 MATLAB/SIMULINK & Coppeliasim

2.4.1 SIMULINK Simscape Multibody

The process of converting a SolidWorks model into Simscape involves several steps that allow the physical components designed in SolidWorks to be utilized within MATLAB/Simulink for dynamic simulations. First, the SolidWorks CAD model is exported using the Simscape Multibody Link plug-in, which generates an XML file and associated physical geometry files. These files are then imported into Simscape using the Simscape Multibody Import tool. This conversion translates the SolidWorks geometrical and mechanical properties, such as masses, inertias, and joint constraints, into equivalent Simscape blocks. Each SolidWorks component, such as links and joints, becomes a Simscape block with defined physical connections. The resulting model is then connected in Simulink with additional control elements, such as motors or sensors, to create a complete simulation environment. As shown in Figure 3.1, the imported Simscape model of a robotic arm consists of multiple rigid bodies (links) and joints, along with motor actuation to drive the movements of the arm.

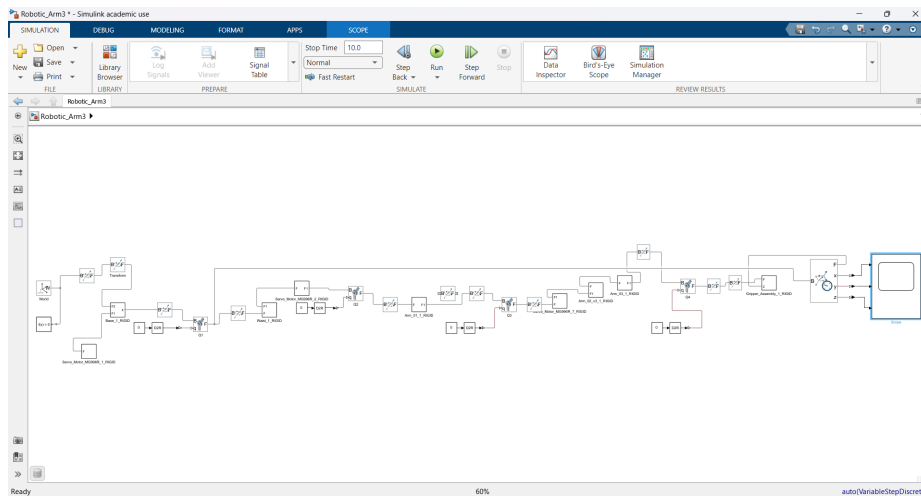


Figure 2.3: Simscape Model of Robotic Arm Imported from SolidWorks

The robotic arm in the image is a 3D model that was originally designed in SolidWorks and then converted to Simscape Mechanics for simulation. Simscape Mechanics is a MATLAB-based simulation environment that allows engineers to model and simulate mechanical systems. By converting the SolidWorks model to Simscape Mechanics, the engineer can analyze the arm's kinematics, dynamics, and control systems. The simulation can help to identify potential issues with the arm's design and optimize its performance before it is built and tested in the real world.

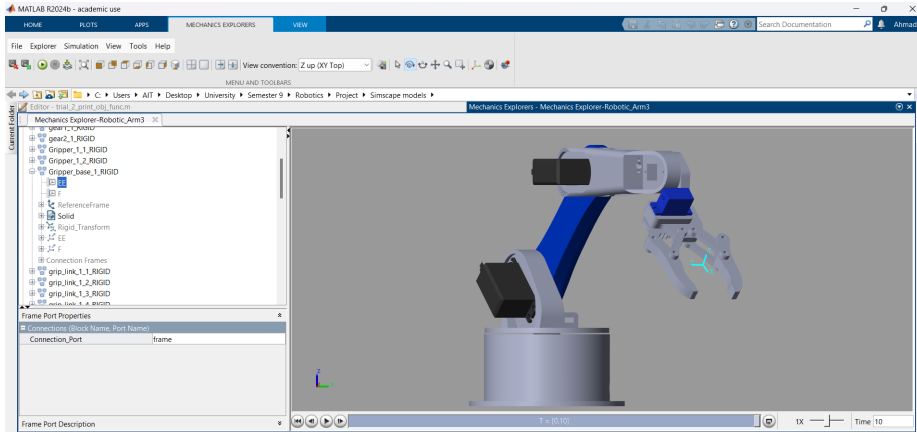


Figure 2.4: Simscape Model of Robotic Arm Imported from SolidWorks

2.4.2 MATLAB Code

The DH convention is a standardized method for representing the position and orientation of each link in a robot using four parameters: the link length, the twist angle, the offset distance, and the joint angle. The code uses these parameters to calculate the homogeneous transformation matrix for each link, which represents the link's position and orientation relative to the base frame. By multiplying the transformation matrices for all the links, the code can determine the position and orientation of the end effector relative to the base frame.

2.4.3 CoppeliaSim

The robotic arm shown in the image is a 3D model that was originally designed in SolidWorks and then exported to CoppeliaSim for simulation. CoppeliaSim is a powerful robotics simulation platform that allows users to test and evaluate the performance of robots in a virtual environment. By exporting the SolidWorks model to CoppeliaSim, we were able to simulate the arm's movements, interactions with its environment, and control systems.

```

Editor - C:\Users\Nabil\Desktop\Semester 9\Robotics\Project\Matlab codes\Main.m
1  transformation_func_Eq.m  forward_kinematics_func.m  forward_kinematics_func_Eq.m  Main.m  transformation_func.m
2  y=forward_kinematics_func_Eq()
3  x=forward_kinematics_func(0,0,0)

Command Window
New to MATLAB? See resources for Getting Started.

>> Main

Y =

12*cos(theta1)*cos(theta2) + 13*cos(theta1)*cos(theta2)*cos(theta3) - 13*cos(theta1)*sin(theta2)*sin(theta3)
12*cos(theta2)*sin(theta1) + 13*cos(theta2)*cos(theta3)*sin(theta1) - 13*sin(theta1)*sin(theta2)*sin(theta3)
11 + 12*sin(theta2) + 13*cos(theta2)*sin(theta3) + 13*cos(theta3)*sin(theta2)

X =

27.4760
-2.6000
13.0560

fx >>

```

Figure 2.5: Matlab code

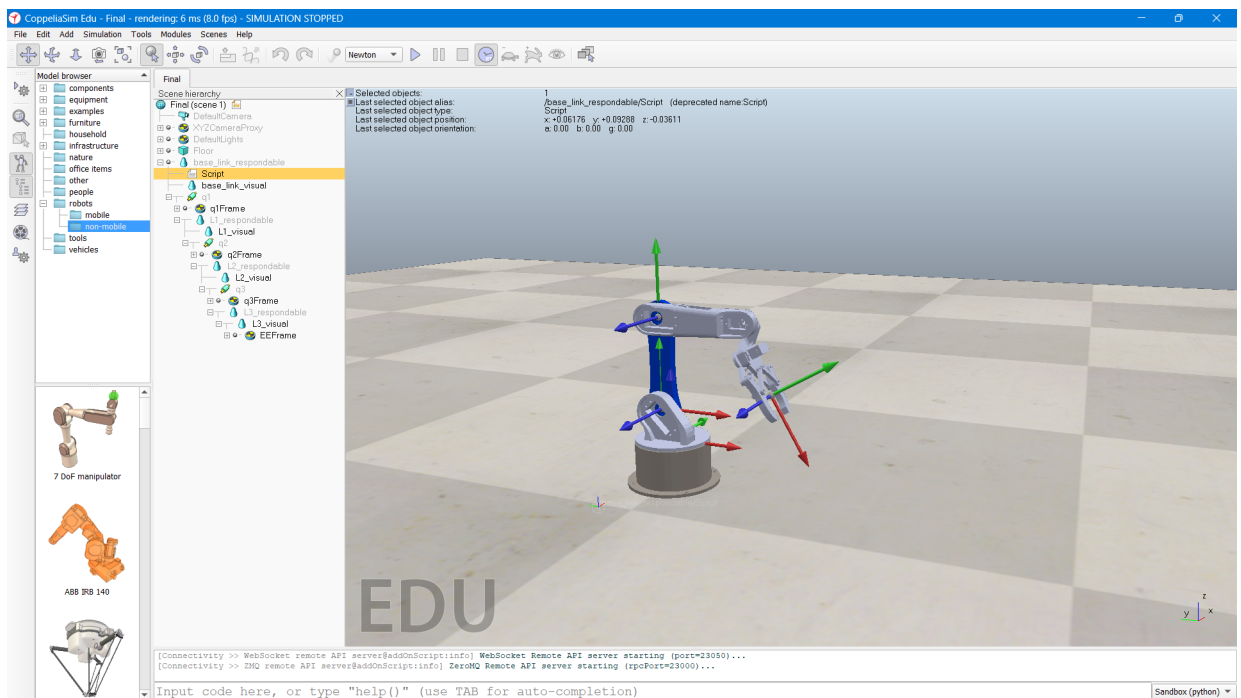


Figure 2.6: Coppeliasim simulation

Chapter 3

Robot Hardware

The hardware of our robotic system was custom-designed and 3D printed, allowing for precise manufacturing of the manipulator's joints, links, and end-effector. The 3D-printed components provide the structure necessary to achieve the required degrees of freedom for movement, while actuators, such as motors, drive these joints to execute specific tasks. Sensors integrated into the system provide real-time feedback on joint position and other environmental factors. A control unit processes this feedback and commands the actuators, ensuring smooth operation. This 3D-printed approach not only allowed for rapid prototyping but also ensured a tailored fit for the robotic design, enabling it to perform designated functions effectively.

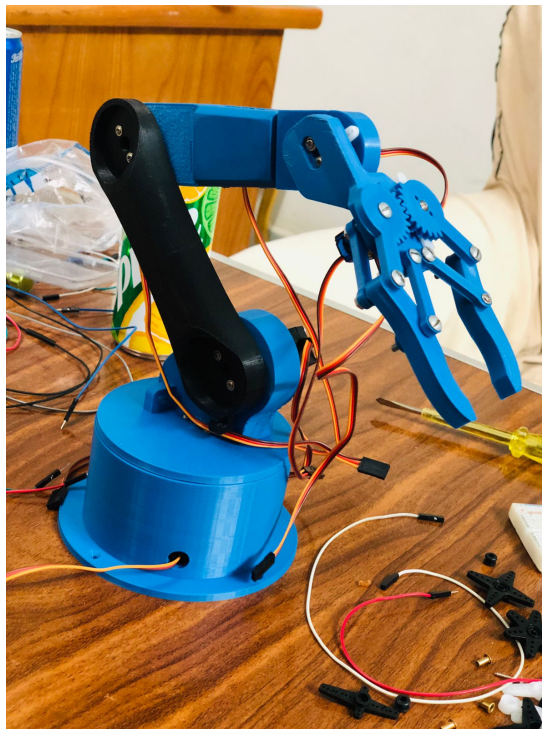


Figure 3.1: 3D printed Robotic Arm